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## DESCRIPTION

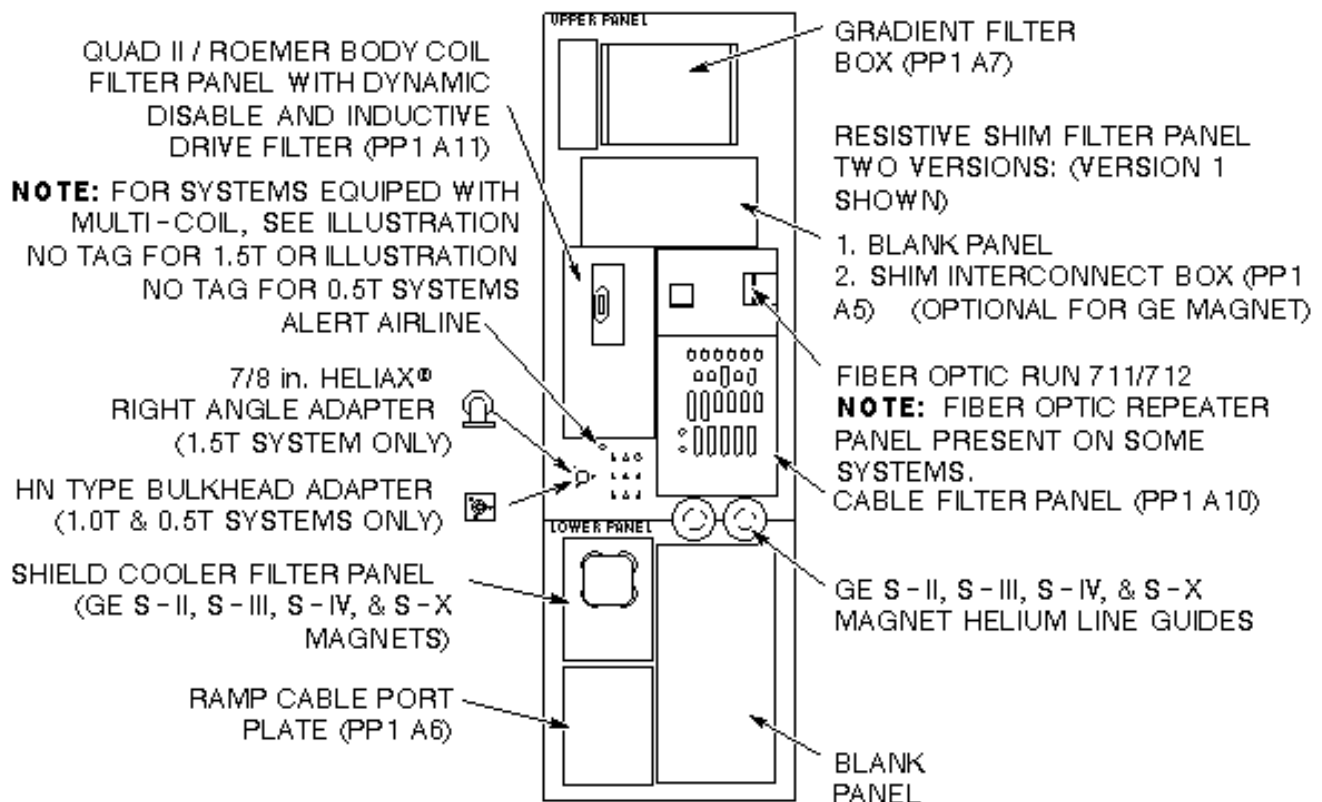
Describes the different version components on the penetration panel and the interconnect run numbers.

## 1- PENETRATION PANEL CONFIGURATIONS

Penetration Panel filter panels in Signa Advantage 1.5T upgrade should be upgraded to one single configuration shown in Illustration L2521A. Systems with 1.5T magnets have a 7/8 in. HELIAX right angle adapter in J44. Systems with 1.0T and 0.5T magnets will have an HN type bulkhead adapter instead.



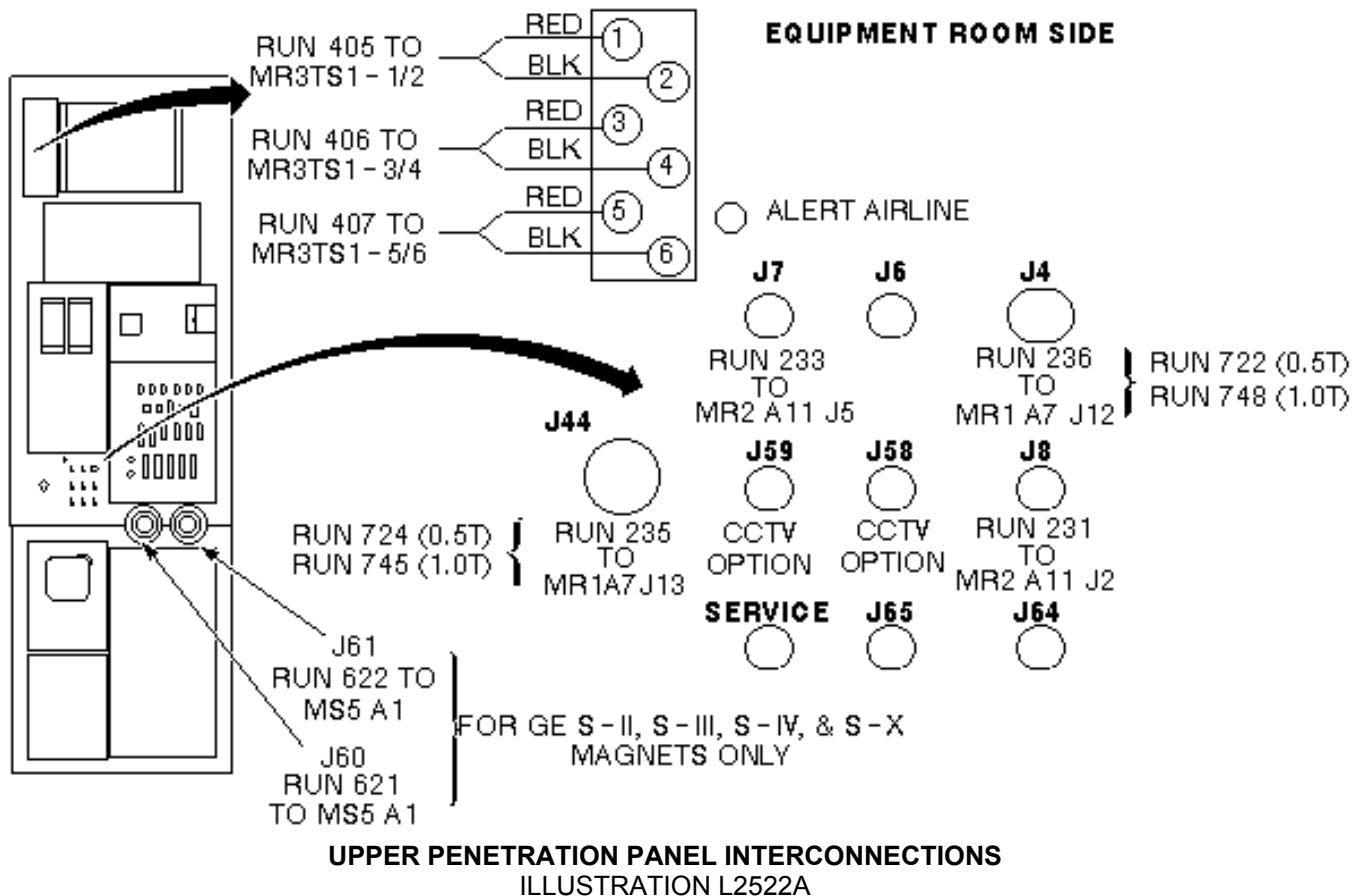
Sites with Oxford Magnet's may have an Ion Pump system connected through the Pen Panel. The Ion Pump consists of two parts, the pump which is on the magnet, and the high voltage supply which is in the scan room. The Ion Pump receives it's power from facility power. Special care must be taken when servicing the Pen Panel on sites that have Ion Pumps because powering down the PDU will not remove power to the Ion Pump.



**SIGNA ADVANTAGE 1.5T, 1.0T, & 0.5T PENETRATION PANEL**  
ILLUSTRATION L2521A

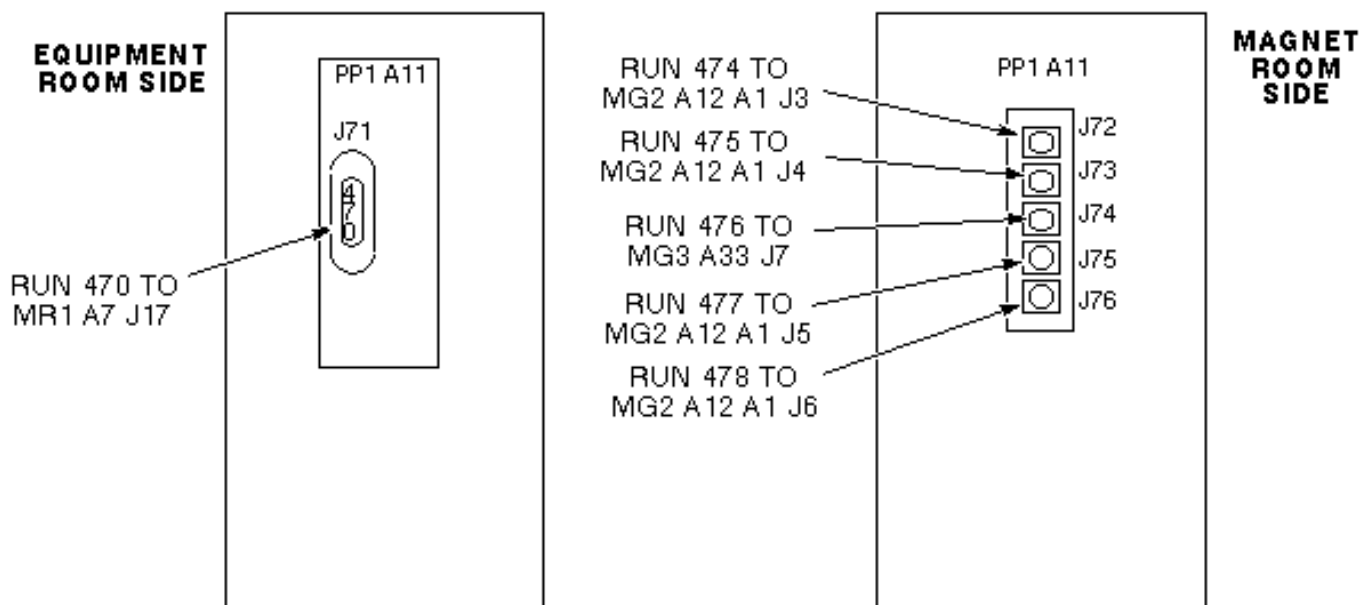
## 2- PENETRATION PANEL CONNECTIONS

Cables connecting to upper Penetration Panel and not connecting within a filter panel are shown in Illustration L2522A. Note different Run Numbers for J4 and J44 for Signa Advantage 1.5T, 1.0T, and 0.5T Systems.



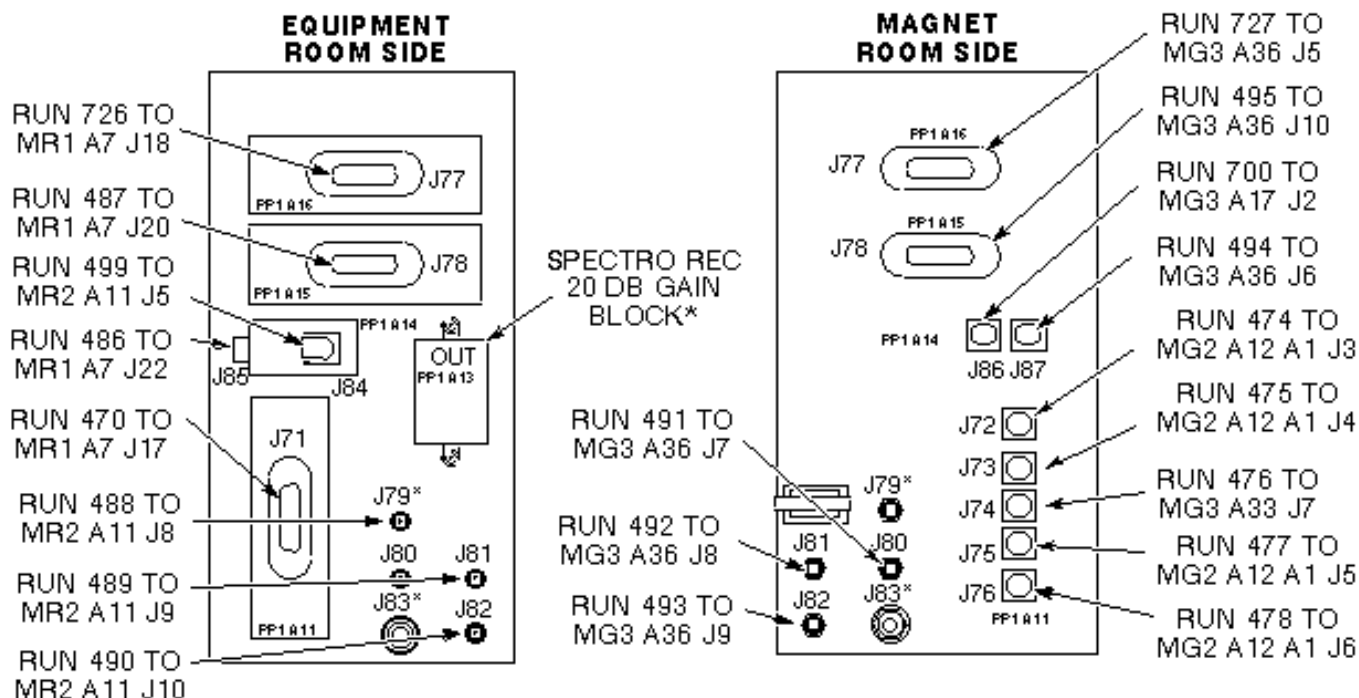
### 3- COIL FILTER PANEL CONNECTIONS

The coil filter panel with a Dynamic Disable and Inductive Drive Filter (PP1 A11) is standard with Signa Advantage 1.5T, 1.0T, and 0.5T systems. (See Illustration L2524A). Upgraded systems will be brought to this configuration. Systems with Multicoil option have a Multicoil Bias Filter (PP1-A16), a Multicoil Switching Filter (PP1-A14), and a Multicoil Receive Select Switch (PP1-A14). (See Illustration L2525A for 1.5T systems with Multicoil option; see Illustration L2526A for 0.5T systems with Multicoil option). Signa Advantage 1.5T systems with Spectroscopy option have a Spectro Receiver 20db Gain Block (PP1 A13). (See Illustration L2525A).



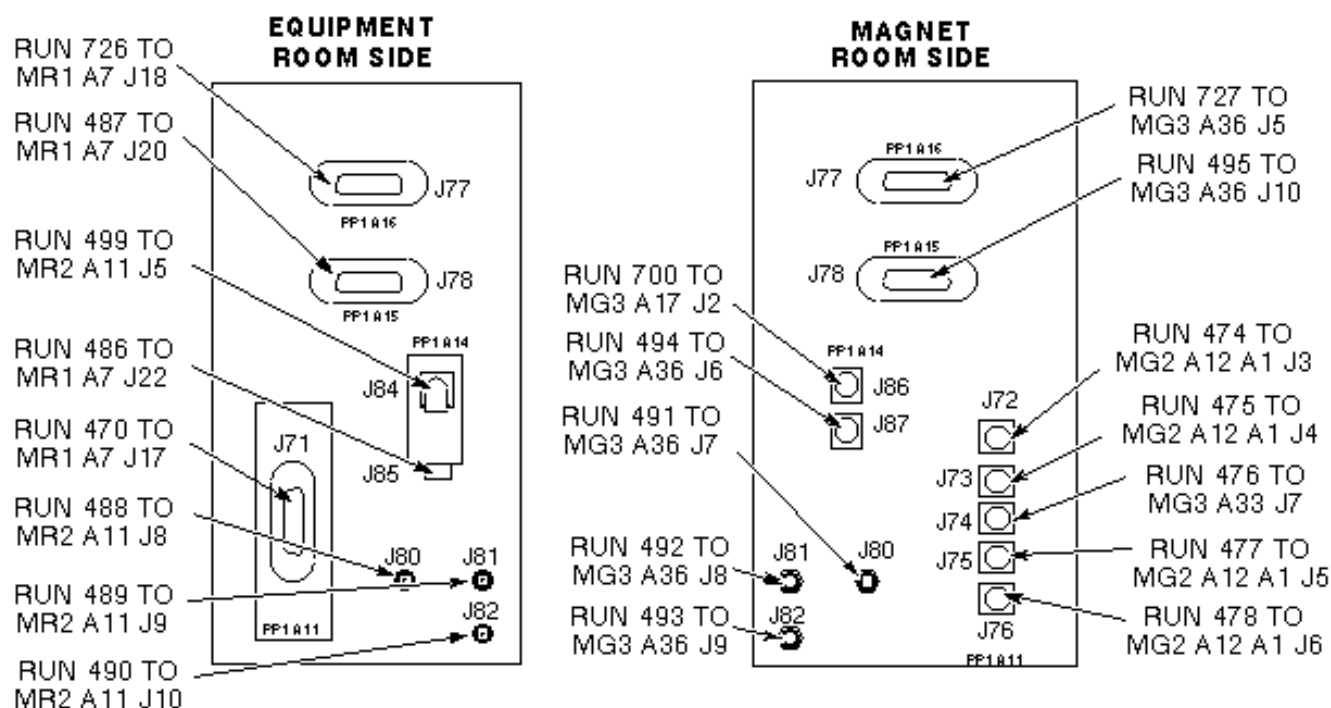
**STANDARD COIL FILTER PANEL**

ILLUSTRATION L2524A



\* For Spectroscopy only

**COIL FILTER PANEL FOR MULTICOIL (1.5T SYSTEMS)**  
ILLUSTRATION L2525A



**COIL FILTER PANEL FOR MULTICOIL (0.5T SYSTEMS)**  
ILLUSTRATION L2526A

#### 4- RESISTIVE SHIM FILTER CONNECTIONS

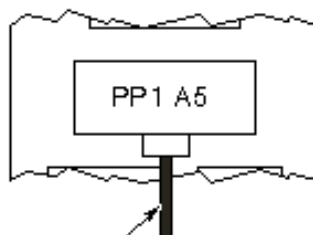
See Illustration L2527A, Resistive Shim Filter Panel.

##### **MAGNET ROOM SIDE**



RUN 607B (GE SI MAGNETS)  
RUN 612 (GE SII/SIII MAGNETS)  
RUN 293 (OXFORD MAGNETS)

##### **EQUIPMENT ROOM SIDE**

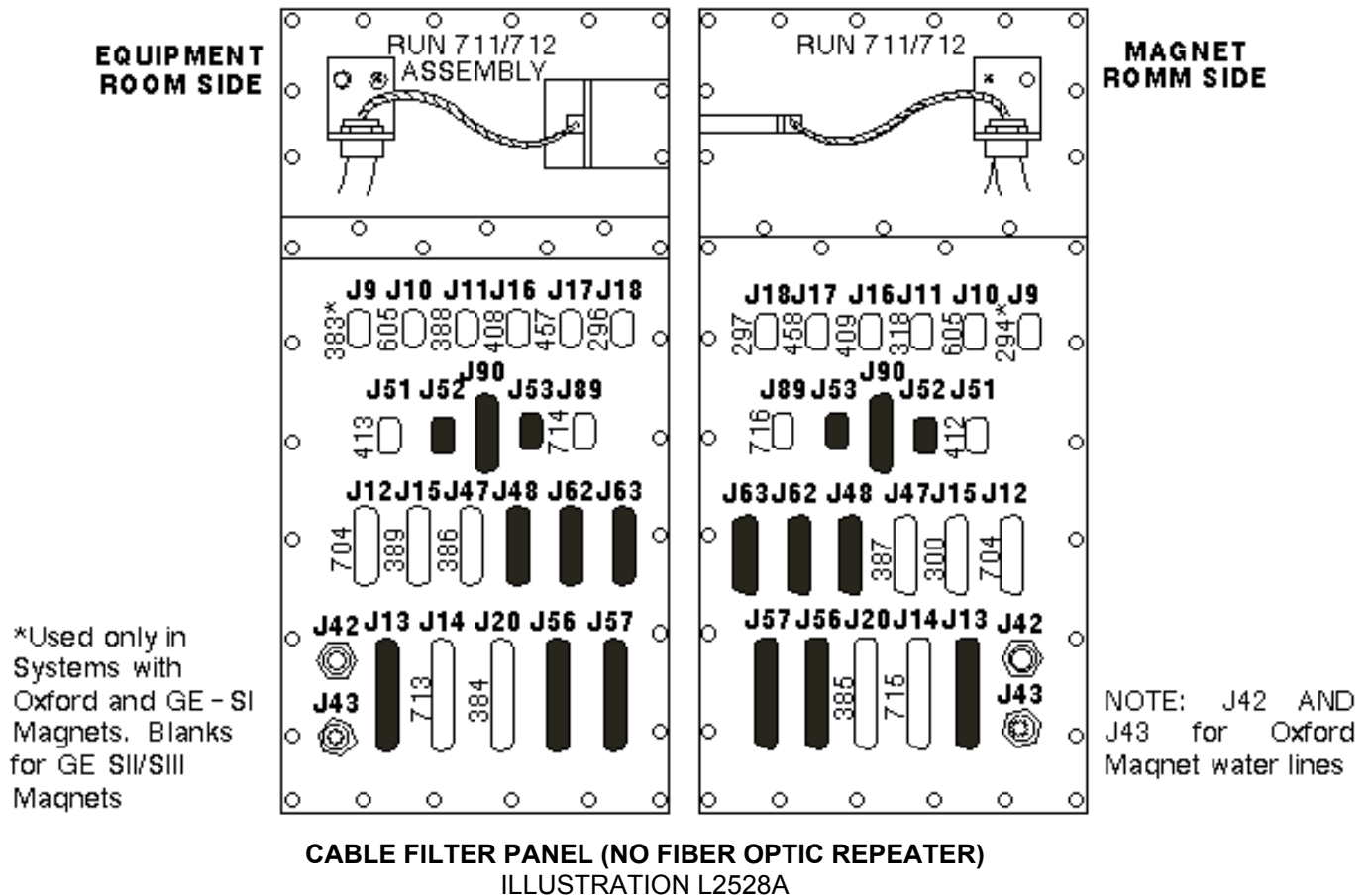


RUN 607A (GE SI MAGNETS)  
RUN 611 (GE SII/SIII MAGNETS)  
RUN 292 (OXFORD MAGNETS)

**RESISTIVE SHIM FILTER PANEL**  
ILLUSTRATION L2527A

## 5- CABLE FILTER PANEL CONNECTIONS

Signa Advantage cables interconnect to cable filter panel as shown in Illustration L2528A (for systems without Fiber Optic Repeater) or Illustration L2529A (for systems with Fiber Optic Receiver). Refer to Table 1 for destinations.



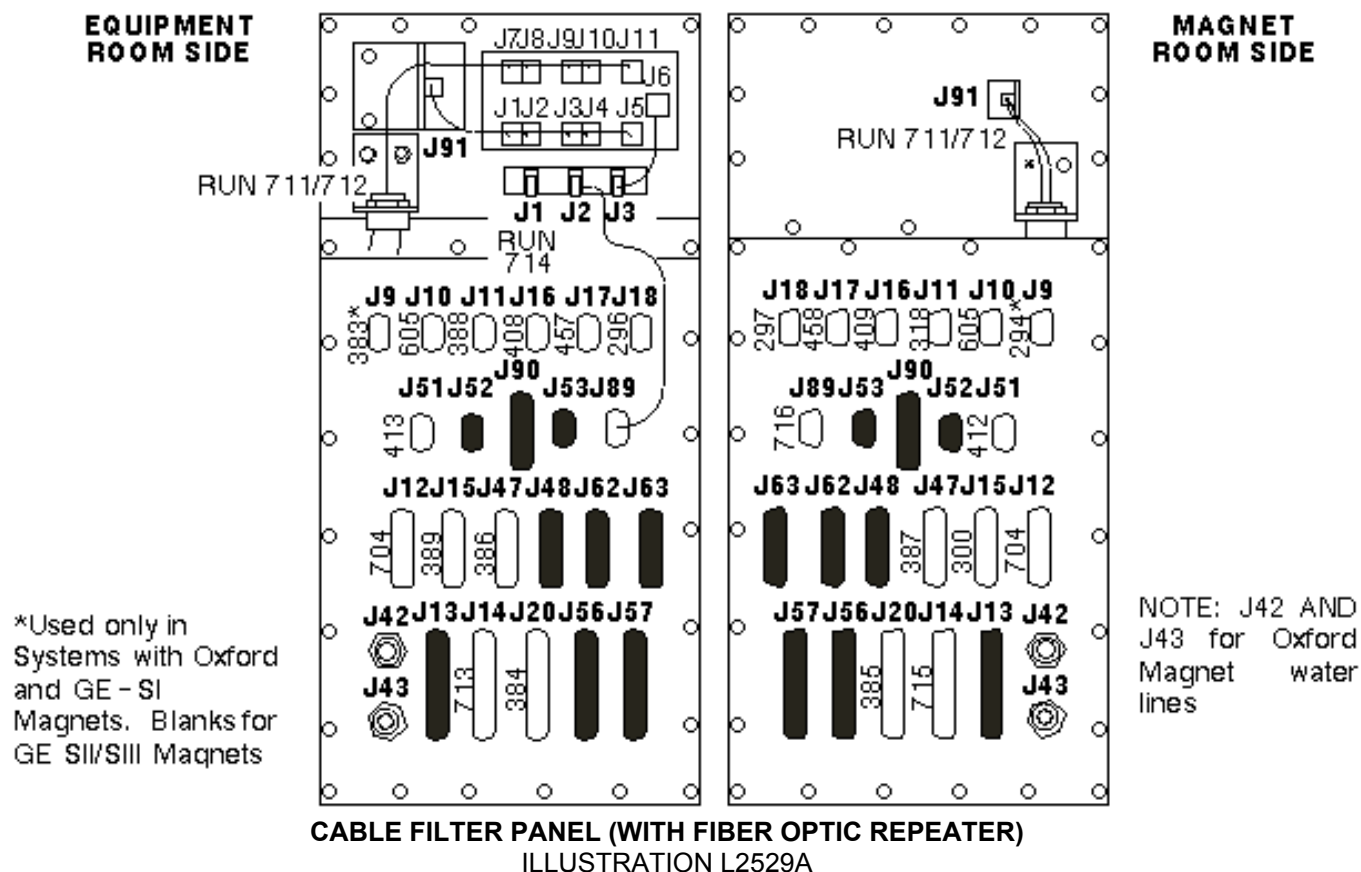


TABLE 1  
CABLE FILTER PANEL INTERCONNECTING CABLES

| PENETRATION PANEL<br>INTERCONNECTING J# | SIGNA ADVANTAGE 1.5T RUNS |                         | SIGNA ADVANTAGE 1.0T & 0.5T RUNS |                      |
|-----------------------------------------|---------------------------|-------------------------|----------------------------------|----------------------|
|                                         | EQUIPMENT<br>ROOM         | MAGNET<br>ROOM          | EQUIPMENT<br>ROOM                | MAGNET<br>ROOM       |
| <b>PP1 J9</b> (9-PIN FILTER)            | BLANK <sup>1</sup>        | BLANK <sup>1</sup>      | BLANK                            | BLANK                |
| <b>PP1 J10</b> (9-PIN FILTER)           | 605 TO MR2 A11 J71        | 605 TO MS1 A3 A11       | 605 TO MR2 A11 J7                | 605 TO MS1 A3 A1     |
| <b>PP1 J11</b> (9-PIN FILTER)           | 388 TO PP2 A15 J22        | 318 TO MG3 A7           | 388 TO PP2 A15 J22               | 318 TO MG3 A7        |
| <b>PP1 J12</b> (25-PIN FILTER)          | 704 TO OC1 A10 J9         | 282 TO MG3 A2 J6        | 704 TO OC1 A10 J9                | 282 TO MG3 A2 J6     |
| <b>PP1 J13</b> (37-PIN CUTOUT)          | BLANK                     | BLANK                   | BLANK                            | BLANK                |
| <b>PP1 J14</b> (37-PIN FILTER)          | 713 TO PP2 A15 J23        | 715 TO MG2 A33 J1       | 713 TO PP2 A15 J23               | 715 TO MG2 A33 J1    |
| <b>PP1 J15</b> (25-PIN FILTER)          | 389 TO PP2 A15 J20        | 300 TO MG2 A15          | 389 TO PP2 A15 J20               | 300 TO MG2 A15       |
| <b>PP1 J16</b> (9-PIN FILTER)           | 408 TO PD1 A1 A3 J3       | 409 TO MG2 A12 A1 J7    | 408 TO PD1 A1 A3 J3              | 409 TO MG2 A12 A1 J7 |
| <b>PP1 J17</b> (9-PIN FILTER)           | 457 TO OM1                | 458 TO OM3              | 457 TO OM1                       | 458 TO OM3           |
| <b>PP1 J18</b> (9-PIN FILTER)           | 296 TO PD1/EO2            | 297 TO EO1              | 296 TO PD1/EO2                   | 297 TO EO1           |
| <b>PP1 J20</b> (37-PIN FILTER)          | 384 TO PP2 A15 J17        | 385 TO MG2 A30 TS1      | 384 TO PP2 A15 J17               | 385 TO MG2 A30 TS1   |
| <b>PP1 J42</b> (BULKHEAD FITTING)       | WATER LINE <sup>2</sup>   | WATER LINE <sup>2</sup> | N/A                              | N/A                  |
| <b>PP1 J43</b> (BULKHEAD FITTING)       | WATER LINE <sup>2</sup>   | WATER LINE <sup>2</sup> | N/A                              | N/A                  |
| <b>PP1 J47</b> (25-PIN FILTER)          | 386 TO PP2 A15 J21        | 387 TO MG2 A29          | 386 TO PP2 A15 J21               | 387 TO MG2 A29       |
| <b>PP1 J48</b> (25-PIN CUTOUT)          | BLANK                     | BLANK                   | BLANK                            | BLANK                |
| <b>PP1 J51</b> (9-PIN FILTER)           | 413 TO PP2 A15 J28        | 412 TO MG3 A32 J1       | 413 TO PP2 A15 J28               | 412 TO MG3 A32 J1    |

Signa Advantage 1.5T, 1.0T, & 0.5T systems, and Signa Advantage1.5T Upgrade Cable Filter Panel interconnecting cables are shown in Table 1above.

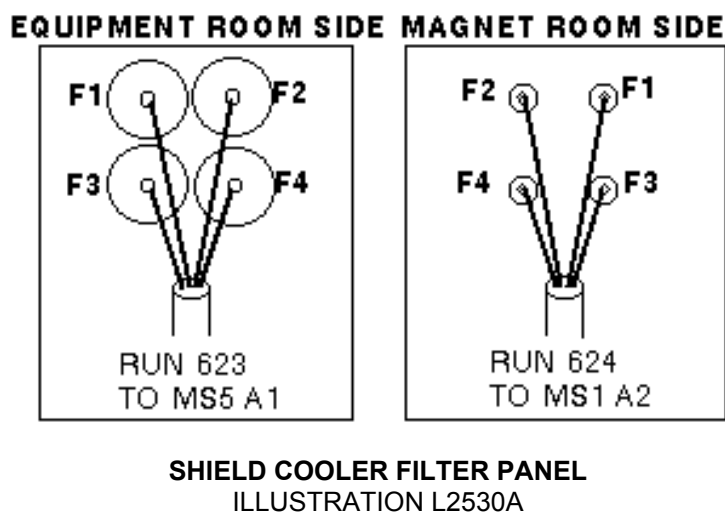
Signa Advantage Upgrade runs that interconnect according to magnet type are listed in Table 2.

TABLE 2  
UPGRADE CABLE FILTER PANEL CONNECTIONS ACCORDING TO MAGNET

| PENETRATION PANEL<br>INTERCONNECTING J#,<br>DEVICE, & FUNCTION | OXFORD MAGNET                 |                                | GE S-I MAGNET             |                   | GE S-II OR GE S-III MAGNET |                         |
|----------------------------------------------------------------|-------------------------------|--------------------------------|---------------------------|-------------------|----------------------------|-------------------------|
|                                                                | EQUIPMENT<br>ROOM             | MAGNET<br>ROOM                 | EQUIPMENT<br>ROOM         | MAGNET<br>ROOM    | EQUIPMENT<br>ROOM          | MAGNET<br>ROOM          |
| <b>J9</b> (9-PIN SUN-D FILTER)<br>NITROGEN METER               | RUN 383 TO<br>MR2 A11 J13     | RUN 294 TO<br>MG1 A2 A1 J1     | RUN 383 TO<br>MR2 A11 J15 | RUN 608 TO<br>MS1 | NOT USED                   | NOT USED                |
| <b>J10</b> (9-PIN SUN-D FILTER)<br>HELIUM METER                | RUN 291 TO<br>MR2 A11 J7      | RUN 295 TO<br>MG1 A1 A2 J1     | RUN 291 TO<br>MR2 A11 J7  | RUN 608 TO<br>MS1 | RUN 605 TO<br>MR2 A11 J7   | RUN 605 TO<br>MS1 A3 A1 |
| <b>J42</b><br>SHIM COIL WATER                                  | WATER LINE TO<br>WATER SUPPLY | WATER LINE TO<br>SHIM (MG1 A1) | NOT USED                  | NOT USED          | NOT USED                   | NOT USED                |
| <b>J43</b><br>SHIM COIL WATER                                  | WATER LINE TO<br>WATER SUPPLY | WATER LINE TO<br>SHIM (MG1 A1) | NOT USED                  | NOT USED          | NOT USED                   | NOT USED                |
| <b>F1, F2, F3, F4</b> (FILTERS)<br>SHIELD COOLER               | NOT USED                      | NOT USED                       | NOT USED                  | NOT USED          | RUN 623 TO<br>MS5 A1       | RUN 624 TO<br>MS1 A2    |

## 6- SHIELD COOLER FILTER PANEL CONNECTIONS

Signa Advantage 1.5T, 1.0T, & 0.5T, and Signa Advantage 1.5T Upgrade Penetration Panels include a filter panel in the lower section for mounting the GE S-II, S-III, S-IV, or S-X Magnet shield cooler filters. Illustration L2530A shows the interconnect to this panel.



## 7- MAGNET RAMP PORT PANEL (PP1 A6)

Signa Advantage 1.5T, 1.0T, & 0.5T and Signa Advantage 1.5T Upgrade Penetration Panels include a removable panel in the lower section. Unfastening captive thumb nuts allows quick removal for feed through of Oxford or GE Magnet ramping cables. This port is designed to accept all cable combinations, according to magnet type, as listed in Table 3.

TABLE 3  
MAGNET RAMP PORT PANEL CABLES

| CABLE FUNCTION       | OXFORD MAGNET        |                         | GE S-I MAGNET            |                          | GE S-II, S-III, S-IV, & S-X MAGNETS |                         |
|----------------------|----------------------|-------------------------|--------------------------|--------------------------|-------------------------------------|-------------------------|
|                      | EQUIPMENT ROOM       | MAGNET ROOM             | EQUIPMENT ROOM           | MAGNET ROOM              | EQUIPMENT ROOM                      | MAGNET ROOM             |
| MAGNET FIELD CURRENT | RUN 288<br>TO MG5    | RUN 288<br>TO MG1 A4 A4 | RUN 601<br>TO MS2 A1     | RUN 601<br>TO MS2 A4     | RUNS 601, 602<br>TO MS6             | RUNS 601, 602<br>TO MS  |
| MAGNET RAMP CONTROL  | RUN 316<br>TO MG5    | RUN 316<br>TO MG2 A13   | RUN 602/605<br>TO MS2 A1 | RUN 602/605<br>TO MS2 A4 | RUN 604<br>TO MS6 & MS7             | RUN 604<br>MS1 A3 A1    |
| SUPERCONDUCTOR SHIM  | NOT USED<br>NOT USED | NOT USED<br>NOT USED    | RUN 603/604<br>TO MS2 A1 | RUN 603/604<br>TO MS2 A4 | RUN 603<br>TO MS7                   | RUN 603<br>TO MS3 A3 A1 |

## REVISION HISTORY

| REV | DATE          | AUTHOR        | PRIMARY REASONS FOR CHANGE                |
|-----|---------------|---------------|-------------------------------------------|
| 0   | July 14, 1998 | J. Saperstein | Initial conversion from Toolbook to Word. |
| 1   | Nov 16, 1999  | P. Kargard    | Added Ion Pump Caution on page 2.         |
|     |               |               |                                           |
|     |               |               |                                           |
|     |               |               |                                           |